A proof of the conjectured generating function for OEIS A263119

Adrian Perez Fontelles and Gaspard Moulinier, Independent researchers

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1 The conjecture

The Online Encyclopedia of Integer Sequences records A263119 as

> *Number of (n+3) X (1+3) 0..1 arrays with each row divisible by 15 and column not divisible by 15, read as a binary number with top and left being the most significant bits*

and carries in its Formula section the line:

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Empirical g.f.:  x*(14 + x + x^2 + x^3 - 14*x^4) / ((1 - x)*(1 + x)*(1
- 2*x)*(1 + x^2)).  - Colin Barker, Jan 01 2019
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The line quoted above is contributed by Colin Barker on Jan 01 2019.

The word *Empirical* — equivalently *Conjecture* — is the entry's own: the statement is recorded as fitted to the published terms, and no proof is given there. As of revision 11, last modified Apr 12 2019, the entry records no proof and links to none, so the statement is open. This note settles it.

Written out, the claim is

$$\sum_{n\geq 1} a(n)\,x^n \;=\; \frac{14x + x^2 + x^3 + x^4 - 14x^5}{1 - 2x - x^4 + 2x^5}.$$

2 The object

The name is read as follows. The reading is not a guess: it is pinned against the entry's own published terms, and against an independent enumeration, before anything is proved (Section 7).

Let A be an $n' \times 4$ array over $\{0, \dots, 1\}$ with $n' = n + 3$ rows. Each row is read as a base-2 number with its leftmost entry the most significant digit, and each column as a base-2 number with its topmost entry the most significant digit.

The condition is that every row is divisible by 15 and every column is not divisible by 15.

$a(n)$ is the number of such arrays. The entry's offset is 1, so its term of index n is the count for n rows.

3 The count is a walk count

Reading a column downwards multiplies its value so far by 2 and adds the new digit, so the residue of a column modulo 15 after i rows determines its residue after $i+1$ rows once the new digit is known. Take as state the vector of the 4 column residues modulo 15; there are at most 15^4 of them.

The rows that may be appended are exactly those whose own value is divisible by 15 — a condition on the row alone — and a state is accepting when every column residue is nonzero.

An array of n' rows is then exactly a walk of length n' from the all-zero state to an accepting state, and every array arises from exactly one walk, so the count is $w^{\mathsf{T}} M^{n'} f$ and is C-finite.

4 The degree bound, derived

The reachable part of the digraph has 15 states, 15 after trimming; identifying states with equal numbers of completions of every length, and then the mirror identification on the edges coming in, leaves $S = 5$ blocks. The count satisfies the linear recurrence given by the characteristic polynomial of the resulting 5×5 matrix. That bound is computed, not assumed.

5 The decision procedure

Let $c_1, \ldots, c_D$ be the coefficients of a claimed recurrence $a(n) = \sum_{i=1}^{D} c_i\, a(n-i)$, let $q(t) = t^D - \sum_{i=1}^{D} c_i t^{D-i}$ be its characteristic polynomial, and put

$$u_m \;=\; \tilde{w}^{\mathsf{T}} L^{\,m-1} q(L)\, \tilde{f} \qquad (m \geq 1).$$

Expanding the powers of L and using the displayed formula for a, one gets $u_m = a(n) - \sum_{i=1}^{D} c_i\, a(n-i)$ with $n = m + D$. So u_m is exactly the residual of the claim at that index, and the recurrence holds at an index n if and only if the corresponding u_m vanishes.

Now $u_m = \tilde{w}^{\mathsf{T}} L^{m-1} y$ with $y = q(L)\tilde{f}$ a fixed vector, so (u_m) satisfies the characteristic polynomial of L, of degree 5: writing that polynomial as $t^5 - \sum_{i=1}^{5} \lambda_i t^{5-i}$ gives $u_{m+5} = \sum_{i=1}^{5} \lambda_i u_{m+5-i}$. Hence 5 consecutive vanishing residuals force every later residual to vanish, by induction. The test is therefore finite; and because it locates the *last* nonvanishing residual, it pins the threshold exactly rather than merely bounding it.

Everything is carried out in exact integer arithmetic on the vector $L^{m-1}y$. The matrix M is never formed.

6 Results

Theorem 1. *For A263119, the generating function is exactly* $\sum_{n \geq 1} a(n)x^n = \frac{14x + x^2 + x^3 + x^4 - 14x^5}{1 - 2x - x^4 + 2x^5}$.

Proof. Let $N(x)$ and $D(x)$ be the numerator and denominator above, $D(0) = 1$. The residual test applied to the recurrence read off D — namely $a(n) = \sum_i c_i a(n-i)$ with c_i the negated coefficients of D — gives vanishing residuals throughout, so the power series $F(x) = \sum_{n \geq 1} a(n)x^n$ satisfies $[x^k](F(x)D(x)) = 0$ for every $k > 6$. The remaining coefficients of $F(x)D(x)$, for $k \leq 13$, were computed from the walk count and agree with $N(x)$ term by term. Hence $F(x)D(x) = N(x)$ identically, which is the assertion. $\square$

7 Verification

Four checks were run, each reproducible from the code accompanying this note, and the first two are what pin the reading of the entry's English.

- **The published terms.** The walk count reproduces all 29 terms the entry publishes, exactly, in integer arithmetic, with the index taken from the entry's stated offset (1) rather than fitted. The first of them are 14, 29, 59, 119, 238, 477, 955, 1911, ...
- **An independent enumeration.** For $n = 1, \ldots, 4$ every one of the 2^{4n} arrays was written out and tested cell by cell against the condition as the entry states it — no transfer matrix, no row window, a separate piece of code. The counts agree with the walk count and with the entry.
- **The claim on the raw data.** Each claim was evaluated on the entry's published terms alone, with no model involved, wherever the proved range and the published data overlap. It holds there.
- **The annihilation test.** Carried to the derived bound $S = 5$ over the whole vector, in exact integer arithmetic, never sampled and never truncated.

8 References

1. OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A263119, revision 11, last modified Apr 12 2019.
2. R. P. Stanley, *Enumerative Combinatorics*, Vol. 1, 2nd ed., Cambridge University Press, 2012; §4.7, the transfer-matrix method.
3. M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011; C-finite sequences and their closure properties.
4. J. Berstel and C. Reutenauer, *Noncommutative Rational Series with Applications*, Cambridge University Press, 2011; linear representations of counting automata and their reduction.